Cataloguing a Library's Book Damage with Computer Vision: Progress Report II

SUMMARY: YOLO to identify books, Damage model identifies damage (yes/no), and HTML file displays which books are damaged; currently labeling more images and retraining model for better results & text identification.

Current Progress:

1. Ran dataset of images through Yolo and cropped each image classified as a book
2. Labeled each of these images as "damaged" or "not damaged" (or skipped if image wasn't actually a book)
3. Trained the model on these images and labels (put all data into a csv file and turned into a tensor object to train)
4. Built an HTML file/script that shows the user which books are damaged and which aren't.

User Workflow:

1. User takes a picture of a bookshelf with multiple books (Fig.1)
2. YOLO will identify individual books (Fig.2)
3. The damage model will classify each as damaged or not
4. The individual, cropped images of the books and their corresponding labels are displayed in the html file on a browser (Fig.3).



Figure 1: An example of an image taken by a user.



Figure 2: An example of a cropped image YOLO identified.

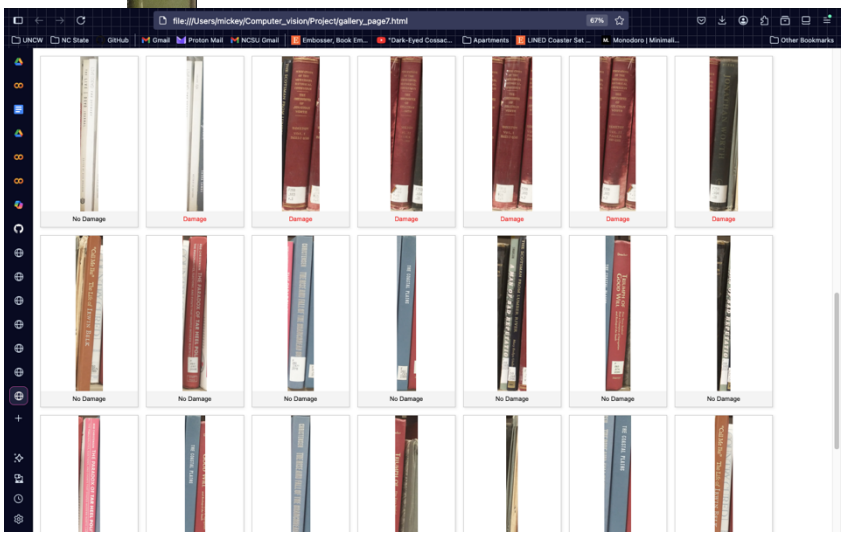


Figure 3: An example of what the html file looks like to the user. Note that damaged books have red text to highlight that they require care from the librarian.

Proposed Scope Changes

1. Dataset does not include images of the front cover of books.
   a. The reasoning for this was that it would be better to build the original model with only one type of book (spine or front) to reduce potential confusion.
   b. Because the idea behind this project is that a librarian could snap a picture of a shelf and get damage labels for all the books on the shelf, I chose to only do spines.
2. I wanted to do text identification so that I could have the book title, author, and published year as info beneath each book in the HTML page. Yet, I'm having trouble finding a library that performs well on the data (or how I need to preprocess the data).
   a. Pytesseract doesn't perform well at all (both with original spine, rotated spine, or gaussian-blurred-canny-edges images), but I've heard of an Amazon model that does well identifying text, so I want to try that.
3. I also wanted to plug in the GPT API to identify similar books for each, but I will not have time to do this because of the text-identification issue.
   a. I definitely want to do this sometime, but it might not be for this project (I will have a lot of time this summer).
4. Also, I wanted to make this a mobile app with Flask and have a SQL backend, but this will also have to be pushed to this summer due to time constraints.
   a. I want to focus on strengthening my damage detection model and trying the text identification.